

Abhayrajsinh Parmar

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Education

Bachelor of Engineering
New LJ Institute of Engineering and Technology

Oct. 2020 – June 2024
Ahmedabad, India

Technical Skills

Languages:	Python, JavaScript, SQL
AI & LLM:	RAG (Retrieval-Augmented Generation), Generative AI, Prompt Engineering, LLM Agents, Model Context Protocol (MCP), LLM Evaluation
LLMs & Serving:	OpenAI API, vLLM (self-hosted open-weight models), Hugging Face
Retrieval & Vector Search:	Qdrant, Hybrid Search (Dense + BM25), Embeddings, LangChain
Backend & APIs:	FastAPI (Async), REST APIs, Node.js / Express
Databases:	MongoDB, PostgreSQL, MySQL
Data & Documents:	PyMuPDF, Tesseract OCR, BeautifulSoup, Pandas, NumPy
DevOps & Cloud:	Docker, AWS, Git / GitHub

Experience

- AI/ML Engineer** | Maven Profcon Services LLP, Ahmedabad, Gujarat, India Aug 2025 – Sept 2026
- Built and shipped production LLM / Generative-AI features end-to-end on an asynchronous FastAPI backend, serving both OpenAI and self-hosted open-weight models (via vLLM) with token streaming and auth middleware.
 - Developed shared AI infrastructure reused across services: a Qdrant vector layer with hybrid (dense + BM25) retrieval, an embedding service, and asyncio semaphore-based concurrency for fair multi-user throughput.
 - Researched and benchmarked open-weight LLMs for self-hosting — weighing GPU / server cost, quantization, active vs. total parameters (MoE), context-window length and throughput — to select cost-effective models served via vLLM.
 - Applied production practices — system-prompt design, Pydantic output validation, exponential-backoff retries, structured logging, and Dockerized deployment on AWS.
- Backend Developer** | ADS Vizion, Ahmedabad, Gujarat, India Jun 2024 – Jul 2025
- Developed and maintained accounting and business software using Node.js, Express and MySQL, building backend features for invoicing, e-invoice / e-GST generation, financial reporting and transaction management.
 - Built a shipping-management module covering load / unload jobs, party and supplier masters, cargo and vehicle details, report generation and role-based access (admin / godown).

Projects

- PDF Summarization & Chatbot** | FastAPI · Qdrant · OpenAI · RAG · OCR
- Built a document Q&A chatbot with hybrid RAG — dense semantic search (Qdrant) fused with sparse BM25 ranking, run concurrently via asyncio.gather and deduplicated before prompting the LLM.
 - Engineered OCR ingestion (PyMuPDF + Tesseract) with LangChain recursive chunking, per-chunk embedding bounded by an asyncio semaphore, and TTL-cached BM25 indexes to avoid rebuilds on repeat queries.
 - Streamed token-by-token responses over FastAPI (StreamingResponse) with per-document chat history persisted in MongoDB.
- Regulatory Chatbot with MCP** | FastMCP · MCP · Qdrant · OpenAI
- Built a regulatory chatbot that gives the LLM tool access through a Model Context Protocol server (FastMCP, streamable-HTTP) consumed via an MCP ClientSession.
 - Implemented the retrieval stack — chunking, embedding, Qdrant search, medical-device classification, content extraction, and a query combiner that rewrites follow-ups into standalone queries for multi-turn context.
 - Grounded answers strictly in retrieved chunks and persisted conversation history to reduce hallucination.
- Resume Ranking Engine** | FastAPI · OpenAI · MongoDB · Pydantic
- Built an LLM-driven engine scoring resumes against a JD across weighted skills, experience and education, with weight profiles per candidate level (intern / fresher / experienced).
 - Combined batched LLM calls for must-have / good-to-have skill matching and experience extraction with deterministic scoring for degree level, recency and employment gaps.
 - Modeled JD and scoring schemas with Pydantic and exposed JD management and ranking through REST APIs.
- WhatsApp AI Agent** | FastAPI · Qdrant · OpenAI · OCR
- Built a multimodal WhatsApp assistant that ingests text, PDFs and images (Tesseract OCR) and merges them into one structured context bundle per turn.
 - Added LLM query rewriting for follow-ups, retrieval-grounded responses, rolling conversation summarization, and per-user history in MongoDB.
- USFDA Document Pipeline** | asyncio · vLLM · PyMuPDF · Tesseract · Qdrant · MongoDB
- Built a high-throughput pipeline that scrapes USFDA filings, extracts text with PyMuPDF + Tesseract OCR fallback, and indexes embeddings into Qdrant.
 - Ran a self-hosted open-weight LLM via vLLM with bounded async concurrency (semaphores + CPU thread pool), exponential-backoff retries and per-document timeouts for reliable large-batch runs.
 - Used MongoDB bulk upserts (UpdateOne) for idempotent writes and tracked per-document status for failure recovery and reprocessing.